

UNIVERSITY COLLEGE OF JAFFNA

UNIVERSITY OF VOCATIONAL TECHNOLOGY

HIGHER NATIONAL DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY

NATIONAL VOCATIONAL QUALIFICATION LEVEL 06

REQUIREMENTS DOCUMENTATION

Online Food Ordering and Management System

Software Change Management (K72C001M12)

Assessment No 01

Varshika Kalanithy
JF/ICT/24/10

REQUIREMENTS DOCUMENTATION

Online Food Ordering and Management System

Document Version: 1.0
Document Status: Baseline
Project Type: Web-Based System

1. Document Information

Item	Details
Project Name	Online Food Ordering and Management System
Document Name	Requirements Documentation
Version	1.0
Status	Baseline
Project Type	Web-Based System
Main Users	Customer, Administrator

2. Introduction

2.1 Purpose

The Online Food Ordering and Management System is a web-based system for ordering and managing food.

Customers can view food items, search the menu, add food items to a cart, place orders, provide delivery details, select a payment method, and check order status.

Restaurant staff can manage food items, prices, availability, and customer orders.

The administrator can manage users and system access.

3. Problem Statement

Many small and medium restaurants receive orders through phone calls, social media messages, or directly from customers.

These methods can cause problems such as:

- Wrong food orders
- Wrong quantities
- Missing customer details
- Wrong delivery information
- Difficulty managing many orders
- Delays in processing orders
- Difficulty checking order status
- Difficulty keeping food prices updated
- Difficulty checking food availability

The proposed system provides one web-based system for customers and restaurants to manage food ordering.

4. Objectives

The main objectives of the system are:

1. Allow customers to view available food items.
2. Allow customers to search for food.
3. Allow customers to view food details and prices.
4. Allow customers to add food items to a cart.
5. Allow customers to change or remove cart items.
6. Allow customers to place food orders.
7. Allow customers to provide delivery details.
8. Allow customers to select a payment method.
9. Allow customers to check order status.
10. Allow restaurant staff to manage food items.
11. Allow restaurant staff to manage prices.
12. Allow restaurant staff to manage food availability.
13. Allow restaurant staff to manage customer orders.
14. Allow administrators to manage users and access.
15. Reduce mistakes in the food ordering process.

5. Scope

5.1 In-Scope

The system will include:

- Customer registration
- Customer login
- Customer account management
- Food item display
- Food search
- Food categories
- Food details
- Food price display
- Food availability
- Shopping cart
- Cart quantity management
- Order placement
- Delivery details
- Cash on Delivery
- Online payment
- Order confirmation
- Order status
- Order history
- Restaurant food management
- Restaurant order management
- Administrator management
- User management
- Input validation
- Role-based access

5.2 Out-of-Scope

The following features are not included in the system:

- GPS tracking
- Live delivery-person tracking
- Real-time customer location tracking

The customer will enter the delivery details manually.

6. Stakeholders

Stakeholder	Responsibility
Customer	View food and place orders
Administrator	Manage users and system access
Project Team	Develop, test, and maintain the system
Restaurant	Prepare and manage food orders

7. User Roles

7.1 Customer

The customer can:

- Register an account
- Login
- View food items
- Search food
- View food details
- Add food to cart
- Change food quantity
- Remove food from cart
- Enter delivery details
- Select payment method
- Place orders
- View order details
- Check order status
- View previous orders
- Manage account details

7.2 Restaurant Staff

Restaurant staff can:

- Login

- Add food items
- Edit food items
- Delete food items
- Update food prices
- Update food availability
- View customer orders
- View order details
- Update order status

7.3 Administrator

The administrator can:

- Login
- View users
- Manage users
- Manage user roles
- Manage user access
- View orders
- Manage system information

8. Functional Requirements

Functional requirements describe what the system must do.

FR-01: Customer Registration

The system shall allow customers to create an account.

The customer shall enter required information such as:

- Name
- Email
- Phone number
- Password
- Delivery information

The system shall check the entered information before creating the account.

FR-02: Customer Login

The system shall allow registered users to login.

The system shall check the entered username/email and password.

If the login details are incorrect, the system shall display an error message.

FR-03: Food Item Display

The system shall display available food items.

Each food item shall contain:

- Food name
- Food image
- Description
- Price
- Category
- Availability

FR-04: Food Search

The system shall allow customers to search for food items.

The customer can enter a food name or keyword.

The system shall display matching food items.

If no food is found, the system shall display a suitable message.

FR-05: Food Categories

The system shall allow food items to be grouped into categories.

Examples:

- Rice
- Short Eats
- Pizza
- Burgers
- Drinks
- Desserts

The restaurant can manage the food categories according to the restaurant menu.

FR-06: Food Details

The system shall allow customers to view detailed information about a food item.

The details shall include:

- Food name
- Description
- Price
- Image
- Category
- Availability

FR-07: Add Food to Cart

The customer shall be able to add an available food item to the cart.

The system shall store:

- Food item
- Quantity
- Unit price
- Subtotal

FR-08: Update Cart

The customer shall be able to change the quantity of food items.

The system shall automatically update the subtotal and total amount.

FR-09: Remove Food from Cart

The customer shall be able to remove an item from the cart.

The system shall update the cart total after the item is removed.

FR-10: Order Placement

The customer shall be able to place an order.

Before confirming the order, the system shall display:

- Food items
- Quantities
- Prices
- Total amount

- Delivery details
- Payment method

The customer must confirm the order before it is placed.

FR-11: Delivery Details

The customer shall enter the required delivery information.

The system shall save the delivery information with the order.

The system will not use GPS or live location tracking.

FR-12: Payment Method

The system shall allow customers to select a payment method.

The payment methods are:

1. Cash on Delivery
2. Online Payment

The selected payment method shall be stored with the order.

FR-13: Order Confirmation

After an order is successfully placed, the system shall display an order confirmation.

The system shall provide the customer with the order details.

FR-14: Order Status

The system shall allow customers to check their order status.

Possible statuses include:

- Pending
- Confirmed
- Preparing
- Ready
- Delivered
- Cancelled

Restaurant staff shall update the order status.

FR-15: Order History

The system shall allow customers to view their previous orders.

The order history shall contain:

- Order ID
- Order date
- Food items
- Quantity
- Total amount
- Payment method
- Order status

9. Restaurant Management Requirements

FR-16: Add Food Item

Restaurant staff shall be able to add new food items.

The required information can include:

- Food name
- Description
- Price
- Category
- Image
- Availability

FR-17: Edit Food Item

Restaurant staff shall be able to edit existing food information.

The system shall save the updated information.

FR-18: Delete Food Item

Restaurant staff shall be able to delete food items.

The system should ask for confirmation before deleting an item.

FR-19: Manage Food Availability

Restaurant staff shall be able to change food availability.

For example:

Available or **Unavailable**

Customers shall not be able to order unavailable food.

FR-20: Manage Food Price

Restaurant staff shall be able to update food prices.

The new price shall be displayed to customers.

FR-21: View Customer Orders

Restaurant staff shall be able to view customer orders.

The order information shall include:

- Order ID
- Customer details
- Food items
- Quantity
- Total amount
- Delivery details
- Payment method
- Order status

FR-22: Update Order Status

Restaurant staff shall be able to update the status of an order.

The updated status shall be shown to the customer.

10. Administrator Requirements

FR-23: User Management

The administrator shall be able to view and manage registered users.

FR-24: User Role Management

The administrator shall be able to manage user roles.

The main roles are:

- Customer
- Restaurant Staff
- Administrator

FR-25: User Access Management

The administrator shall be able to control user access.

A user shall only access the functions allowed for their role.

FR-26: View Orders

The administrator shall be able to view system orders.

11. Non-Functional Requirements

Non-functional requirements describe how the system should work.

NFR-01: Usability

The system should be easy to use.

Buttons, menus, forms, and messages should be clear.

NFR-02: Performance

The system should respond to normal user actions within a reasonable time.

Food lists, cart information, and order information should load without unnecessary delays.

NFR-03: Security

The system shall protect user information.

Passwords shall be stored securely.

Users shall only access functions allowed for their roles.

NFR-04: Data Validation

The system shall check user input before saving data.

Examples:

- Required fields cannot be empty.
- Email must have a valid format.
- Phone number must follow the required format.
- Price must be a valid number.
- Quantity must be greater than zero.

NFR-05: Reliability

The system should correctly save food, customer, and order information.

Order information should not be lost during normal system operation.

NFR-06: Maintainability

The system should be developed using a clear structure so that future changes can be made easily.

NFR-07: Browser Compatibility

The system should work on commonly used modern web browsers.

NFR-08: Access Control

The system shall restrict access according to the user's role.

For example, a customer should not be able to access restaurant management pages.

12. Business Rules

ID	Business Rule
BR-01	A customer must have an account before placing an order.
BR-02	A customer cannot order an unavailable food item.
BR-03	Food quantity must be greater than zero.
BR-04	Food price must be a valid positive value.
BR-05	An order must contain at least one food item.
BR-06	Required delivery information must be entered before placing an order.
BR-07	A payment method must be selected before confirming an order.
BR-08	Only authorized restaurant staff can manage food items.
BR-09	Only authorized restaurant staff can update order status.
BR-10	Only administrators can manage system-level user access.
BR-11	GPS and live delivery tracking are not included.

13. Use Cases

UC-01: Register Account

Actor: Customer

Main Flow

1. Customer opens the registration page.
2. Customer enters the required information.
3. Customer submits the form.
4. System validates the information.
5. System creates the account.
6. System displays a successful registration message.

UC-02: Login

Actor: Customer / Restaurant Staff / Administrator

Main Flow

1. User opens the login page.
2. User enters login details.
3. System checks the details.
4. System identifies the user role.
5. System opens the correct dashboard.

Alternative Flow

If the login details are incorrect:

1. System rejects the login.
2. System displays an error message.

UC-03: Search Food

Actor: Customer

Main Flow

1. Customer opens the food menu.
2. Customer enters a search keyword.
3. System searches food records.
4. System displays matching food items.

UC-04: Add Food to Cart

Actor: Customer

Main Flow

1. Customer selects a food item.
2. Customer selects the quantity.
3. Customer clicks Add to Cart.
4. System checks availability.
5. System adds the item to the cart.
6. System updates the cart total.

UC-05: Place Order

Actor: Customer

Main Flow

1. Customer opens the cart.
2. Customer checks the food items.
3. Customer enters delivery details.
4. Customer selects payment method.
5. System calculates the total.
6. Customer confirms the order.
7. System creates the order.

8. System displays the order confirmation.

UC-06: Manage Food

Actor: Restaurant Staff

Main Flow

1. Restaurant staff logs in.
2. Staff opens food management.
3. Staff adds, edits, or deletes food information.
4. System validates the information.
5. System saves the changes.

UC-07: Manage Order

Actor: Restaurant Staff

Main Flow

1. Staff logs in.
2. Staff opens order management.
3. Staff selects an order.
4. Staff checks the order.
5. Staff updates the order status.
6. System saves the updated status.

UC-08: Check Order Status

Actor: Customer

Main Flow

1. Customer logs in.
2. Customer opens order history.
3. Customer selects an order.
4. System displays the current status.

14. User Stories

Customer

US-01: As a customer, I want to create an account so that I can place food orders.

US-02: As a customer, I want to login so that I can access my account.

US-03: As a customer, I want to search for food so that I can find food easily.

US-04: As a customer, I want to view food details so that I can decide what to order.

US-05: As a customer, I want to add food to a cart so that I can order multiple items.

US-06: As a customer, I want to change food quantity so that I can order the required amount.

US-07: As a customer, I want to remove food from the cart so that I can change my order.

US-08: As a customer, I want to enter delivery details so that the restaurant knows where to deliver my order.

US-09: As a customer, I want to select a payment method so that I can pay for my order.

US-10: As a customer, I want to check my order status so that I know the current state of my order.

Restaurant Staff

US-11: As restaurant staff, I want to add food items so that customers can see the available food.

US-12: As restaurant staff, I want to edit food information so that the food details remain correct.

US-13: As restaurant staff, I want to update food availability so that customers cannot order unavailable food.

US-14: As restaurant staff, I want to view customer orders so that I can prepare the orders.

US-15: As restaurant staff, I want to update order status so that customers can check their order progress.

Administrator

US-16: As an administrator, I want to manage users so that I can control system access.

US-17: As an administrator, I want to manage user roles so that users can access the correct functions.

US-18: As an administrator, I want to view orders so that I can check system order records.

15. Acceptance Criteria

AC-01: Registration

- Customer can open the registration page.
- Customer must enter required information.
- Invalid information must show an error message.
- Valid information must create an account.

AC-02: Food Search

- Customer can enter a search keyword.
- System displays matching food items.
- System displays a message when no food is found.

AC-03: Cart

- Customer can add available food.
- Customer can change quantity.
- Customer can remove food.
- System calculates the total correctly.

AC-04: Order

- Customer must enter required delivery details.
- Customer must select a payment method.
- Order must contain at least one food item.
- System must save the order after confirmation.
- System must display order confirmation.

AC-05: Food Management

- Restaurant staff can add food.

- Restaurant staff can edit food.
- Restaurant staff can delete food.
- Restaurant staff can update availability.
- System saves the changes correctly.

AC-06: Order Status

- Restaurant staff can update order status.
- Customer can view the updated order status.

16. MoSCoW Priority

Must Have

- Customer registration
- Customer login
- Food display
- Food search
- Food details
- Shopping cart
- Order placement
- Delivery details
- Cash on Delivery
- Online payment
- Order confirmation
- Order status
- Food management
- Order management
- User management
- Input validation
- Role-based access

Should Have

- Customer order history
- Customer account management
- Food availability management
- Basic order records

Could Have

- Additional food filters
- Additional reports

- Additional dashboard information

Will Not Have

- GPS tracking
- Live delivery tracking

17. Product Backlog

ID	Requirement	Priority	User
PB-01	Customer registration	Must	Customer
PB-02	Customer login	Must	Customer
PB-03	View food items	Must	Customer
PB-04	Search food	Must	Customer
PB-05	View food details	Must	Customer
PB-06	Add food to cart	Must	Customer
PB-07	Update cart	Must	Customer
PB-08	Place order	Must	Customer
PB-09	Delivery details	Must	Customer
PB-10	Cash on Delivery	Must	Customer
PB-11	Online payment	Must	Customer
PB-12	Order status	Must	Customer
PB-13	Order history	Should	Customer
PB-14	Add food	Must	Restaurant Staff
PB-15	Edit food	Must	Restaurant Staff
PB-16	Delete food	Must	Restaurant Staff
PB-17	Manage availability	Must	Restaurant Staff
PB-18	View orders	Must	Restaurant Staff
PB-19	Update order status	Must	Restaurant Staff
PB-20	User management	Must	Administrator
PB-21	Role management	Must	Administrator

18. Data Requirements

The main data used by the system is described below.

18.1 Customer

The system may store:

- Customer ID

- Name
- Email
- Phone number
- Password
- Address
- Account status

18.2 Food Item

The system may store:

- Food ID
- Food name
- Description
- Price
- Category ID
- Image
- Availability

18.3 Category

The system may store:

- Category ID
- Category name
- Category description
- Status

18.4 Order

The system may store:

- Order ID
- Customer ID
- Order date
- Total amount
- Delivery details
- Payment method
- Order status

18.5 Order Item

The system may store:

- Order Item ID
- Order ID
- Food ID
- Quantity
- Unit price
- Subtotal

18.6 User

The system may store:

- User ID
- Name
- Email
- Password
- Role
- Account status

19. Input Validation Requirements

The system shall validate information before saving it.

19.1 Required Fields

Required fields cannot be empty.

If a required field is empty, the system shall show an error message.

19.2 Email

The email must follow a valid email format.

Example:

```
customer@gmail.com
```

An invalid email should not be accepted.

19.3 Phone Number

The phone number must follow the required format.

Example:

`0771234567`

If an incorrect number is entered, the system shall display an error message.

19.4 Price

Food price must be a valid positive number.

Example:

`850.00`

The system shall not accept:

`ABC` or `-500`

19.5 Quantity

Food quantity must be greater than zero.

Valid:

`1, 2`

Invalid:

`0 ,-1 , ABC`

20. Security Requirements

The system shall:

1. Require users to login before accessing protected functions.
2. Store passwords securely.
3. Prevent unauthorized access.
4. Use role-based access.
5. Validate user input.
6. Protect customer information.

7. Prevent customers from accessing restaurant management functions.
8. Prevent normal users from accessing administrator functions.

21. System Constraints

The system has the following constraints:

1. The system is web-based.
2. Users need a web browser.
3. An internet/network connection is required for online use.
4. GPS tracking is not included.
5. Live delivery tracking is not included.
6. User access depends on the assigned role.
7. Only authorized users can manage restaurant information.
8. Online payment depends on the selected payment service.

22. Assumptions

The system is based on these assumptions:

1. Customers have access to the internet.
2. Customers enter correct delivery information.
3. Customers provide correct personal information.
4. Restaurant staff keep food information updated.
5. Restaurant staff update food availability correctly.
6. Restaurant staff update order status correctly.
7. The restaurant has the required food items when they are marked as available.
8. Online payment services are available when customers choose online payment.

23. Main System Workflow

Customer Workflow

Register / Login

↓

View Food

↓

Search Food

↓

View Food Details

↓

Add to Cart

↓

Check Cart

↓

Enter Delivery Details

↓

Select Payment Method

↓

Confirm Order

↓

Order Created

↓

Restaurant Receives Order

↓

Restaurant Processes Order

↓

Order Status Updated

↓

Customer Checks Order Status

Restaurant Workflow

Login

↓

View Dashboard

↓

Manage Food

↓

View Customer Orders

↓

Process Order

↓

Update Order Status

↓

Order Completed

24. Requirements Traceability Matrix

Requirement	User Story	Acceptance Criteria
FR-01 Registration	US-01	AC-01
FR-02 Login	US-02	AC-01
FR-04 Search Food	US-03	AC-02
FR-06 Food Details	US-04	AC-02
FR-07 Add to Cart	US-05	AC-03
FR-08 Update Cart	US-06	AC-03
FR-09 Remove Cart Item	US-07	AC-03
FR-10 Place Order	US-08	AC-04
FR-11 Delivery Details	US-08	AC-04
FR-12 Payment	US-09	AC-04
FR-14 Order Status	US-10	AC-06
FR-16 Add Food	US-11	AC-05
FR-17 Edit Food	US-12	AC-05
FR-19 Availability	US-13	AC-05
FR-21 View Orders	US-14	AC-05
FR-22 Update Status	US-15	AC-06
FR-23 User Management	US-16	AC-05
FR-24 Role Management	US-17	AC-05

25. Version Control

The requirements document shall use version numbers to record changes.

Version	Description	Status
0.1	Initial requirements	Draft
0.2	Added customer requirements	Draft
0.3	Added restaurant requirements	Draft
0.4	Added administrator requirements	Draft
0.5	Added non-functional requirements	Draft
0.6	Added user stories	Draft
0.7	Added acceptance criteria	Draft
0.8	Added traceability matrix	Draft
0.9	Requirements review	Draft
1.0	Final requirements baseline	Baseline

26. Requirement Change Process

If a requirement needs to be changed after Version 1.0:

1. Identify the requirement.
2. Record the requested change.
3. Give the reason for the change.
4. Check the effect on other requirements.
5. Check the effect on the database.
6. Check the effect on development.
7. Check the effect on testing.
8. Update the requirements document.
9. Update the version number.
10. Record the change.

27. Requirements Review Checklist

Before approving the requirements document:

- ☐ Customer requirements are included.
- ☐ Restaurant requirements are included.
- ☐ Administrator requirements are included.
- ☐ Functional requirements are included.
- ☐ Non-functional requirements are included.
- ☐ Business rules are included.
- ☐ User stories are included.
- ☐ Acceptance criteria are included.
- ☐ Data requirements are included.
- ☐ Validation requirements are included.
- ☐ Security requirements are included.
- ☐ Payment requirements are included.
- ☐ Order status requirements are included.
- ☐ GPS/live tracking is excluded.
- ☐ Requirements are linked to user stories.
- ☐ Requirements have been reviewed.

28. Requirements Baseline

Baseline Version: 1.0

This document represents the requirements baseline for the Online Food Ordering and Management System.

The baseline covers:

- Customer management
- Food management
- Food search
- Shopping cart
- Order management
- Delivery information
- Payment methods
- Order status
- Restaurant management
- Administrator management
- User access
- Data validation
- Security

Changes made after the baseline should follow the requirement change process.

29. Conclusion

The Online Food Ordering and Management System provides a single web-based system for customers and restaurants.

Customers can view and search food items, add food to a cart, place orders, provide delivery information, select a payment method, and check order status.

Restaurant staff can manage food items, prices, availability, and customer orders.

Administrators can manage users and system access.

The system is intended to reduce mistakes and delays that can happen when food orders are handled through phone calls, social media messages, or direct communication.

GPS and live delivery tracking are not included in this project.

This requirements documentation can be used as the main reference for system design, database design, development, and testing.